

KEYNOTE ADDRESS

The Impact of Mathematical Sciences on Nation Building

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Theme: From Theory to Impact — Mathematical Sciences, Optimization and AI in National Development

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All protocols duly observed.

Distinguished ladies and gentlemen, good morning.

Let me thank the organizers of this conference for the honour of standing before you this morning. And let me say at once that your theme this year — From Theory to Impact — is the correct theme. Not merely a fashionable one. The correct one. Because impact is precisely what has been missing, and impact is what we have come here to discuss.

My subject is the impact of the mathematical sciences on nation building. But before we enter that subject, come with me. Let us go back a little.

They Did It by Hand

Geometry. Trigonometry. Calculus. Differential equations. Probability. Logarithms.

I want you to remember something about the people who built these things. They had no computer. They had no calculator. Many of them had no electricity in the room where they worked. What did they have? Paper. Pen. Candlelight. And a stubborn mind.

Let me give you one example, and I want it to enter your spirit.

Johannes Kepler set out to determine the true shape of the orbit of Mars. One planet. Just one. Do you know how long it took him? Years. Something in the region of nine hundred pages of calculation, worked entirely by hand — and if he made one small error in the middle of it, he would return to the beginning and start again.

Years of a man's life. For one planet. And what did he give the world when he finished? The laws of planetary motion — the foundation upon which, three centuries later, we would place a satellite in the sky.

If you were Kepler, would you not have grown weary? Would you not have said, “This thing is too difficult; let somebody else attempt it”? But he did not stop. And because he did not stop, look at what we inherited.

Four Equations, One Modern World

Let me show you what that hand calculation built.

A man named James Clerk Maxwell sat down and wrote four equations. Four. Simply four equations describing how electricity and magnetism behave.

And from those four equations, what emerged? Radio emerged. Television emerged. Radar emerged. Satellites emerged. The mobile telephone in your pocket at this moment — the one you were pressing before you entered this hall — stands upon those four equations.

Consider the rest. The aeroplane: human beings looked at a bird and said, “We want that.” But wanting was not sufficient. It required fluid dynamics and calculus before that heavy iron could leave the ground. Mathematics is what lifted it.

Radioactivity — exponential decay, the half-life. That is how we came to understand the atom, and it is the same mathematics that today treats cancer in a hospital.

Modern medicine: do you know why an MRI can look inside your body without cutting you open? Fourier transforms. Mathematics from the eighteen hundreds is the reason your physician can see your brain today.

Global mobility — trigonometry and orbital mechanics. That is how GPS knows where you are standing. And insurance and pensions — probability. Somebody worked out how to place a number upon risk, and that is how a nation learned to carry the burden of accident and of old age.

Distinguished ladies and gentlemen, everything we call modern life was built upon mathematics done slowly, painfully, by hand.

So What Is Our Excuse?

Permit me now to ask a question, and I would like each of us to answer it quietly, within our own mind.

| *They had nothing, and they built everything. So what is our excuse?*

Because today, the ordinary laptop upon your desk will solve in one second the problem that consumed four years of Kepler's life. One second. The mathematics that Newton could not attempt because his hand would tire — we can now run it ten thousand times before breakfast.

So tell me. Somebody should tell me.

Should our children still be dying from diseases we already know how to cure?

Should our people still be hungry, in a country with this much land, this much rainfall, this much sunlight?

Should our national grid still be collapsing in the year 2026?

Should we still be planning an entire nation upon guesswork?

It cannot be. It should not be.

We are not short of intelligence in this country. We are not short of brilliant minds — half of them are seated in this hall this morning, and the other half we have exported abroad. We are not short of the mathematics, either. The mathematics is already done. It is published. A great deal of it is free.

So what exactly are we doing?

That is why this subject matters. That is the whole question I have come to place before you: can we take the mathematics we already possess, and the tools that exist today, and use them to actually build this nation?

The Bottleneck Has Moved

Here is the most significant shift of our lifetime, and I want you to hear it clearly.

For the whole of human history, mathematics was limited by the human being holding the pencil. You could conceive of a system of ten thousand equations, but you could not solve it, because you would die first. So we simplified. We made every model small enough for a man to hold in his hand.

That limit is finished.

Formerly the question was: can we compute it? The machine was the bottleneck. Today the question is: will we apply it? The bottleneck has moved from the machine to the mind. From the processor to the policy.

So if a nation remains poor and disordered in the year 2026, “the problem is too complex” is no longer an acceptable answer. That excuse has expired. Today it is a choice.

What Is Nation Building?

We say “nation building” every day. In every budget speech. In every campaign. At every commissioning ceremony. The words have worn smooth, like a coin that has passed through too many hands. Let me restore their edges.

Nation building is not concrete. If pouring concrete were sufficient, we would already be a developed nation — for we have poured a great deal of concrete in this country. Some of it twice, in the same place, in the same year.

Nation building is the deliberate construction of the systems, institutions, identity and capabilities by which a people becomes a functioning, self-sustaining and dignified nation.

But we are mathematicians here, so permit me to state it in our own language.

Nation building is the movement of a nation from a local maximum to the global maximum.

And there, ladies and gentlemen, is the entire story of our country in a single sentence.

Because a local maximum is a very comfortable place. From where you stand upon that small hill, every direction appears to be a step downward. Every reform appears to be a loss. Every change appears to be a risk. So what do you do? You remain there. And you give it a respectable name. You call it stability.

But a local maximum is a comfortable place to die.

Nation building is having the courage to descend from that small comfortable hill, so that you may climb the true mountain.

Magnitude Without Direction

And observe one further thing. Nation building is not a scalar. It is a vector.

A scalar possesses magnitude alone. Size. Quantity. We have abundant population, abundant resources, abundant noise. Two hundred million and more. Oil. Gas. Solid minerals. Land. Sunshine. In magnitude we are extremely well provided for.

But a vector possesses magnitude and direction.

Many nations possess magnitude. Very few possess direction. We are a large nation moving very quickly. The only difficulty is that we are not always certain where we are going.

| We have magnitude in abundance. What we lack is direction.

Whose Responsibility Is It?

Let us now ask the uncomfortable question. Whose responsibility is this?

Is it the responsibility of government? In part. But if nation building were solely a function of government, then every country possessing a government would by now be built — and we know that is not so.

Is it the responsibility of the international community? Permit me to be very direct this morning. No nation in the history of this world has ever been developed by foreigners. Foreign assistance is a boundary condition. It is not the differential equation.

Is it the responsibility of our elders? They set the initial conditions — but many of them will not be present for the solution. Is it the responsibility of the young? They will inherit the answer whether or not they were permitted to do the working.

So here is the correct answer, mathematically.

It is a coupled system. Every one of us is a variable sitting inside somebody else's equation. There is no term you may set to zero without altering the entire solution. The teacher. The trader. The customs officer. The professor. The student. The man who inflates the contract. The woman who reports him. Every one of us is a coefficient.

So the question is not whose responsibility it is. The question is: what is your coefficient, and what is its sign? Positive, negative — or, most commonly in this country, zero.

And now permit me to speak candidly to my own profession.

A great deal of Nigerian academic mathematics is a magnificent contribution of exactly zero. Beautiful papers. Rigorously peer-reviewed. Published in reputable journals. Read by no one who has ever signed a budget. We optimised for publication. We did not optimise for impact.

And that is precisely why your theme this year, From Theory to Impact, is not decoration. It is an accusation — and a correct one.

Nigeria Today

Let us look at ourselves honestly for a moment.

The World Bank estimates that ten million additional Nigerians fell into extreme poverty in 2025, bringing us to roughly fifty-one percent of the population below the international poverty line. Half of us.

The International Monetary Fund recorded average inflation of approximately thirty-one percent in 2024, easing to around twenty-four percent by early 2025. Our infrastructure deficit is estimated at more than one hundred billion United States dollars.

And upon the harmonised learning scale, where three hundred represents the bare minimum and six hundred and twenty-five represents advanced attainment, our students score approximately three hundred and nine. We are barely clearing the floor. And where are our children failing most severely? Mathematics. So the very pipeline that ought to produce the people who will repair all of this — that pipeline is itself broken.

Permit me to add one more figure. We have not conducted a census since 2006.

Twenty years of planning an entire country upon an estimate. And everyone in this room knows the first law of applied mathematics: you cannot optimise what you have not measured. We are operating a nation of two hundred million people upon numbers we last verified when some of the students in this hall were in primary school.

A Plan That Cannot Deliver

Now I come to the point at which mathematics ceases to be theory and becomes evidence.

Somebody actually performed the mathematics upon our national plan.

A published study applied multi-objective optimisation to Nigeria's Sustainable Development Goals under Agenda 2030. The result was this: taken individually, the goals were achievable at eighty-one percent, eighty-three percent, and eighty-nine percent. Entirely respectable figures.

But when they were placed together within one system, the overall achievement function could reach only about seventy-three percent.

Do you understand what that is telling us?

It is telling us that the goals are competing against one another for the same money. The plan, as written, cannot deliver what it promises. Not because our leaders lack goodwill. Not because our people are indolent. Because the plan is infeasible.

That is a mathematical statement. Not politics, not opinion — it has no tribe and no party. And it was sitting inside a journal, where those who write our national plans would never encounter it.

We have been writing national plans in the manner of an idle student writing a proof. He begins by saying: assume the result.

The Structure Is the Body; Mathematics Is the Nervous System

Let us then speak of structures, for we are not the first people to attempt nation building. Functioning countries possess structures. And I wish to show you the mathematics concealed within each one.

First, the national statistical system. This is the eye of the state — sampling, estimation, error bounds. Without it, government is a driver behind a painted windscreen.

Second, planning and budgeting. This is constrained optimisation and nothing else. Maximise welfare, subject to revenue, subject to debt, subject to time. Every budget in this world is a linear programme. Most of ours are written as wish lists.

Third, security. Graph theory. Network centrality. Because you do not defeat a network with a convoy. You defeat it by identifying the nodes that hold it together, and removing them.

Fourth, justice. Queueing theory. A case backlog is a queue, and in any queue where the arrival rate exceeds the service rate, that queue runs to infinity. So justice delayed is not merely justice denied. It is a queue that has become mathematically unstable, and it will never recover unaided.

Fifth, public health. Epidemiological modelling. The basic reproduction number. Cold chain logistics. Every life saved in a pandemic is saved twice — once by a physician, and once, earlier, by a differential equation.

Sixth, energy. Load balancing. Economic dispatch. Our grid collapses are not acts of providence. They are unsolved optimisation problems, collapsing in public.

Sixteen Sectors, One Foundation

And it does not end there. Consider the full range.

Security. Peace. Justice. Governance. Health. Education. Planning. Logistics. Employment. Innovation. Agriculture. Food security. Entertainment. Energy. Anti-corruption. Finance.

Sixteen sectors. And beneath every single one of them, the same discipline.

Agriculture — we lose a substantial portion of our harvest after we have already grown it. That is not a farming failure. That is a routing and storage failure. And routing is mathematics.

Entertainment — yes, even entertainment. Why is the world now listening to Nigerian music? Part of it is our talent. But part of it is a recommendation algorithm upon a streaming platform, and that algorithm is somebody's matrix factorisation. Our greatest cultural export is riding upon mathematics that we neither built nor control.

Sixteen sectors, one foundation. That is not coincidence. That is the definition of a foundational science.

Mathematics Catches Liars

Let me offer you one that may raise a smile.

Corruption. We treat it as a moral problem, and it is. We treat it as a legal problem, and it is. But mathematically, corruption is a problem of anomaly detection.

Consider Benford's Law. In genuine numerical data — real invoices, real populations, real river lengths — the leading digit one appears approximately thirty percent of the time. Not eleven percent, as one might expect. Thirty.

Now, when a man sits down to invent numbers, he does not know this. He distributes his digits neatly, because he believes that is what randomness looks like. Human beings are exceedingly poor at simulating randomness.

So an inflated contract carries a statistical fingerprint — visible before anyone confesses, before anyone testifies, before anyone is arrested.

Mathematics catches liars. And this is the finest part: mathematics cannot be negotiated with. You cannot telephone mathematics at night and plead with it.

The Slow Bleeding

So what becomes of a nation that does not respect this?

Nothing dramatic. And that, distinguished colleagues, is precisely the danger.

When you ignore physics, a bridge falls down. Everyone sees it. Everyone discusses it. A panel of inquiry is constituted.

When you ignore mathematics, nothing collapses at all. You simply become — quietly, expensively, permanently — suboptimal.

You construct the road along the wrong corridor. You site the hospital where the people are not. You order too much of the drug that will expire, and too little of the one that would have saved somebody's mother. You fight last year's insurgency upon this year's terrain. You produce graduates for occupations that no longer exist.

No one is arrested. No headline is written. The country simply operates at sixty percent of what it might have been — permanently — and we give it a name. We call it our peculiar national challenges.

The cost of ignoring mathematics is never a collapse. It is a slow bleeding.

And there is a second consequence that is more dangerous still. When decisions are not founded upon evidence, that space does not remain empty. Something must occupy it. And what occupies it? Sentiment. Ethnicity. Political convenience. Whoever spoke last and loudest in the meeting.

Because mathematics is the only argument in the room that has no tribe. Remove it, and you have removed the sole referee that cannot be bought.

But let me caution my own profession also, that we do not grow proud. Bad mathematics is worse than no mathematics, for it carries an authority it has not earned. Remember the statistician who drowned crossing a river of average depth three feet. On average, he was perfectly well. And remember always that correlation does not imply causation — though I must concede that it does raise its eyebrow suggestively.

Six Things We Must Do

Diagnosis without prescription is merely elegant complaining. So permit me to offer six things. Practical things.

First. Change what we reward in academic mathematics. Measure every department not solely by papers published, but by problems solved for a real Nigerian institution. Adopt a ministry. Adopt a local government. Adopt a teaching hospital. From theory to impact is not a slogan for a banner. It ought to be a key performance indicator.

Second. Place mathematicians in the room where the problem is being formulated — not where the answer is being justified. Because in optimisation, the formulation determines the solution. Permit politicians to formulate while mathematicians merely compute, and you will obtain a highly efficient answer to the wrong question.

Third. Repair the pipeline at its source. Our children are not failing mathematics because they are dull. They are failing because we teach it as a ritual — memorise, reproduce, forget. A child who has never once used mathematics to answer a question he himself cared about has not been taught mathematics. He has been taught obedience.

Fourth. Build the data. Conduct the census. Digitise the registries. Open the data. Every capable thing we might construct is presently starving at the input layer.

Fifth. Demand quantitative literacy of leadership. Not calculus — merely the capacity to examine a model and ask: what is the assumption here, and what is the margin of error upon that number? A leader who cannot ask that will be captured by whoever brings the most confident presentation.

Sixth. Build the bridge to deployment. This is our work at RAIN — taking the theory and placing it into a robot, an embedded system, a model running upon a device somebody can hold in his hand. Mathematics that cannot be deployed is merely a beautiful idea. Mathematics that has been deployed is a nation.

What We Would See

And let me describe what it would actually look like, so that it does not remain abstract.

In health, medicines would arrive before they are needed rather than afterwards. Outbreaks would be flattened by prediction rather than by panic. Specialists would be posted according to disease burden, not according to proximity to the capital.

In education, teachers would be posted where the children actually are — not where the posting is convenient.

In agriculture, planting decisions would follow yield models and satellite data, post-harvest losses would fall, and food prices would be settled by forecasting rather than by decree.

In energy, a grid that balances rather than a grid that collapses. And in governance, budgets that are feasible before they are signed, and inflated contracts flagged automatically before the money departs the account.

And hear this well: every single item upon that list is already a solved problem somewhere else in the world today. The mathematics exists. It is published. A great deal of it is open source and free.

| *We are not waiting for a discovery. We are waiting for a decision.*

Artificial Intelligence Is Mathematics

Which brings me to where we now stand. The future is not approaching. The future has entered the room and taken a seat.

Artificial intelligence is already within our medicine, our agriculture, our banking, our classrooms, our security, our courts, our music, our elections.

And I want this hall to hear the next sentence very clearly, for it is the most important thing I shall say this morning.

| *Artificial intelligence is not a computing revolution. It is a mathematics revolution wearing a computing costume.*

Consider what lies within the machine. Linear algebra. Multivariable calculus. Probability. Optimisation. Information theory. That is all. That is the entire contents.

So everybody seated in this hall is already an artificial intelligence researcher. The subject into which the whole world is now pouring trillions of dollars is your own subject. You were engaged in it before it became fashionable.

And that is why the opportunity before us is extraordinary. This country was excluded from the industrial revolution by capital. We were excluded from the digital revolution by infrastructure. But this one? Its principal raw material is mathematical talent — and of that we have a great deal: young, hungry, and seated in this room.

The Other Edge of the Blade

But every opportunity carries its own danger. Those who wish us harm possess the same mathematics we possess. Very often, they possess better funding.

Consider steganography — the science of concealing information within something that appears innocent. Recent research demonstrates that AI-generated images can now conceal a

payload in a manner that is, by the information-theoretic definition, essentially perfect. Statistically, you cannot distinguish that image from any ordinary image. And there are documented cases of malicious code hidden within images posted openly upon social media.

So understand this: an image is no longer merely an image. It can be a message. It can be a weapon.

Consider synthetic media. A voice can be cloned from thirty seconds of audio. Thirty seconds. In a country where a single false statement can ignite three states before noon, a fabricated recording of a leader is not a technology problem. It is a national security problem.

Consider encryption. Every classified communication, every banking transaction, every citizen record is protected by a single mathematical assumption regarding the difficulty of factorising a large number. Quantum computing is advancing upon that assumption, and other nations have already begun migrating to post-quantum cryptography. So permit me to put a serious question to this hall: where is our own migration plan? Who in Nigeria is working upon this? Because if the answer is no one, then our national secrets carry an expiry date, and somebody else has set it.

So I say to you: we cannot be a nation that merely consumes this mathematics. We must audit it, build it, and defend against it. That work is cryptography, adversarial machine learning, steganalysis, formal verification. It is not glamorous. But it is the difference between digital sovereignty and digital tenancy.

The One Variable We Control

Distinguished ladies and gentlemen, permit me to bring this home.

We have spoken of large matters this morning. Poverty. Insecurity. A national plan that is mathematically infeasible. Quantum computing. Large problems — and it is easy to sit down and feel small.

But there is one variable within this entire system that you and I control completely.

It is the number of people in this nation willing to carry rigour out of the classroom and apply it to the country.

It is the professor who adopts a ministry. It is the postgraduate student who builds the model for the local government, and not solely for the journal. It is the young person at the back of this hall who will spend the coming decade ensuring that Nigeria's decisions can be defended with evidence.

And here is the finest part. For that variable you require no permission. No legislation. No budget line. No one need sign anything on your behalf.

You may begin on Monday morning.

In Closing

We began this morning by remembering people who possessed no computer, no calculator, no electricity — and who nevertheless gave the world the aeroplane, electricity, modern medicine, and the satellite.

We then agreed that today the limit is no longer the machine. Today the limit is ourselves. Today it is a choice.

So let it be recorded that upon this day, in this hall, a gathering of Nigerian mathematicians resolved that From Theory to Impact was not a theme printed upon a banner.

It was an instruction.

A nation is not built by those who admire the problem. A nation is built by those who solve it.

Thank you very much.

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